HOMEWORK 6

Ans 1) a) $$\frac{1}{c^2}\frac{\partial^2\psi(r,t)}{\partial t^2} = \nabla^2\psi(r,t) - \left(\frac{m_e c}{\hbar}\right)^2\psi(r,t)$$

$$\Rightarrow \frac{1}{c^2}\psi^*\frac{\partial^2\psi(r,t)}{dt^2} = \psi^*\nabla^2\psi(r,t) - \left(\frac{m_e c}{\hbar}\right)^2\psi^*\psi$$

also $$\frac{1}{c^2}\frac{\partial^2\psi^*}{\partial t^2} = \nabla^2\psi^* - \left(\frac{m_e c}{\hbar}\right)^2\psi^*$$

$$\Rightarrow \frac{1}{c^2}\psi\frac{\partial^2\psi^*}{\partial t^2} = \psi\nabla^2\psi^* - \left(\frac{m_e c}{\hbar}\right)^2\psi\psi^*$$

$$\Rightarrow \frac{1}{c^2}\left(\psi^*\frac{\partial^2\psi}{\partial t^2} - \psi\frac{\partial^2\psi^*}{\partial t^2}\right) = \left(\psi^*\nabla^2\psi - \psi\nabla^2\psi^*\right)$$

$$\Rightarrow \frac{1}{c^2}\frac{\partial}{\partial t}\left(\psi^*\frac{\partial\psi}{\partial t} - \psi\frac{\partial\psi^*}{\partial t}\right) = \bar{\nabla}\cdot\left(\psi^*\nabla\psi - \psi\bar{\nabla}\psi^*\right)$$

$$\Rightarrow \frac{i\hbar}{2m_e c^2}\frac{\partial}{\partial t}\left(\psi^*\frac{\partial\psi}{\partial t} - \psi\frac{\partial\psi^*}{\partial t}\right) + \bar{\nabla}\cdot\left[\frac{-i\hbar}{2m_e}\left(\psi^*\nabla\psi - \psi\nabla\psi^*\right)\right] = 0$$

$$P(\bar{r},t) = \frac{i\hbar}{2m_e c^2}\left[\psi^*\frac{\partial\psi}{\partial t} - \psi\frac{\partial\psi^*}{\partial t}\right]$$

$$\bar{J}(\bar{r},t) = \frac{\hbar}{2im_e}\left[\psi^*\nabla\psi - \psi\nabla\psi^*\right]$$

$$\Rightarrow \boxed{\frac{\partial P}{\partial t} + \bar{\nabla}\cdot\bar{J} = 0}$$

b) $\psi(\bar{r},t) = A\exp\{i(\bar{p}\cdot\bar{r} - Et)/\hbar\}$

$$\frac{\partial\psi}{\partial t} = \frac{-iE}{\hbar}\psi \quad \text{and} \quad \frac{\partial\psi^*}{\partial t} = \frac{iE}{\hbar}\psi^*$$

$$\rho(\bar{r},t) = \frac{i\hbar}{2m_ec^2}\left[\psi^*\left(\frac{-iE}{\hbar}\psi\right) - \psi\left(\frac{iE}{\hbar}\psi^*\right)\right]$$

$$\rho(\bar{r},t) = \frac{i\hbar}{2m_ec^2}\left(\frac{-2iE}{\hbar}|\psi|^2\right) = \frac{E}{m_ec^2}|\psi|^2$$

For $E < 0$, $\rho < 0$ as $|\psi|^2 > 0$

but when $\psi(r,t) \in \mathbb{R} \Rightarrow \psi^* = \psi$

$$\rho(\bar{r},t) = \frac{i\hbar}{2m_ec^2}\left[\psi\frac{\partial\psi}{\partial t} - \psi\frac{\partial\psi}{\partial t}\right] = 0$$

c) No, we can not ~~def~~ define a standard single particle directly from a single component K-G eq^n.

Ans2) a) $\alpha_j = \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix}$, $\beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$

(a) $\alpha_j^2 = I_4$, $\beta^2 = I_4$

$$\alpha_j^2 = \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} = \begin{pmatrix} \sigma_j^2 & 0 \\ 0 & \sigma_j^2 \end{pmatrix} \Rightarrow \begin{pmatrix} I_2 & 0 \\ 0 & I_2 \end{pmatrix}$$

$$\beta^2 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}\begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} = I_4$$

b) $\{\alpha_i, \alpha_j\}$

$$\alpha_i\alpha_j = \begin{pmatrix}0&\sigma_i\\\sigma_i&0\end{pmatrix}\begin{pmatrix}0&\sigma_j\\\sigma_j&0\end{pmatrix} = \begin{pmatrix}\sigma_i\sigma_j&0\\0&\sigma_i\sigma_j\end{pmatrix}$$

$$\{\alpha_i,\alpha_j\} = \begin{pmatrix}\{\sigma_i,\sigma_j\}&0\\0&\{\sigma_i,\sigma_j\}\end{pmatrix} = \begin{pmatrix}2\delta_{ij}I & \\ 0 & 2\end{pmatrix}$$

$$= \boxed{2\delta_{ij} I_4}$$

$\{\alpha_i,\beta\}$: $\alpha_i\beta = \begin{pmatrix}0&\sigma_i\\\sigma_i&0\end{pmatrix}\begin{pmatrix}I&0\\0&-I\end{pmatrix} = \begin{pmatrix}0&-\sigma_i\\\sigma_i&0\end{pmatrix}$

$$\beta\alpha_i = \begin{pmatrix}I&0\\0&-I\end{pmatrix}\begin{pmatrix}0&\sigma_i\\\sigma_i&0\end{pmatrix} = \begin{pmatrix}0&\sigma_i\\-\sigma_i&0\end{pmatrix}$$

$$\{\alpha_i,\beta\} = \begin{pmatrix}0&-\sigma_i+\sigma_i\\\sigma_i-\sigma_i&0\end{pmatrix} = \boxed{0}$$

c) $\alpha_j\alpha_k = i\epsilon_{jkl}\begin{pmatrix}\sigma_l&0\\0&\sigma_l\end{pmatrix}$

$$\alpha_j\alpha_k = \begin{pmatrix}\sigma_j\sigma_k&0\\0&\sigma_j\sigma_k\end{pmatrix}, \quad \sigma_j\sigma_k = \delta_{jk}I + i\epsilon_j$$

$$\alpha_j\alpha_k = \begin{pmatrix}\delta_{jk}I+i\epsilon_{jkl}\sigma_l & 0\\0&\delta_{jk}I+i\epsilon_{jkl}\sigma_l\end{pmatrix}$$

$$\alpha_j\alpha_k = \delta_{jk}I_4 + i\epsilon_{jkl}\begin{pmatrix}\sigma_l&0\\0&\sigma_l\end{pmatrix}$$

b) $\{\alpha_i, \alpha_j\}$

$$\alpha_i \alpha_j = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} = \begin{pmatrix} \sigma_i \sigma_j & 0 \\ 0 & \sigma_i \sigma_j \end{pmatrix}$$

$$\{\alpha_i, \alpha_j\} = \begin{pmatrix} \{\sigma_i, \sigma_j\} & 0 \\ 0 & \{\sigma_i, \sigma_j\} \end{pmatrix} = \begin{pmatrix} 2\delta_{ij} I & 0 \\ 0 & 2\delta_{ij} I \end{pmatrix}$$

$$= \boxed{2\delta_{ij} I_4}$$

$\{\alpha_i, \beta\}$:

$$\alpha_i \beta = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}\begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = \begin{pmatrix} 0 & -\sigma_i \\ \sigma_i & 0 \end{pmatrix}$$

$$\beta \alpha_i = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}\begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}$$

$$\{\alpha_i, \beta\} = \begin{pmatrix} 0 & -\sigma_i + \sigma_i \\ \sigma_i - \sigma_i & 0 \end{pmatrix} = \boxed{0}$$

c) $\alpha_j \alpha_k = i\epsilon_{jkl} \begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix}$

$$\alpha_j \alpha_k = \begin{pmatrix} \sigma_j \sigma_k & 0 \\ 0 & \sigma_j \sigma_k \end{pmatrix}, \quad \sigma_j \sigma_k = \delta_{jk} I + i\epsilon_{jkl}\sigma_l$$

$$\alpha_j \alpha_k = \begin{pmatrix} \delta_{jk} I + i\epsilon_{jkl}\sigma_l & 0 \\ 0 & \delta_{jk} I + i\epsilon_{jkl}\sigma_l \end{pmatrix}$$

$$\alpha_j \alpha_k = \delta_{jk} I_4 + i\epsilon_{jkl} \begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix}$$

$$= i\epsilon_{jkl} \begin{pmatrix} \sigma_l & 0 \\ 0 & \sigma_l \end{pmatrix}$$

Ans3) $i\hbar \dfrac{\partial \psi(\bar{r},t)}{\partial t} = H\psi(r,t)$

$H = c\alpha\cdot p + \beta m_e c^2$

(a) $\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}$, $\beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$

$p = -i\hbar\nabla = (-i\hbar\,\partial_x, -i\hbar\,\partial_y, -i\hbar\,\partial_z)$

$\bar{\alpha}\cdot\bar{p} = \alpha_x P_x + \alpha_y P_y + \alpha_z P_z = \begin{pmatrix} 0 & \bar{\sigma}\cdot\bar{p} \\ \bar{\sigma}\cdot\bar{p} & 0 \end{pmatrix}$

$\bar{\sigma}\cdot\bar{p} = \begin{bmatrix} P_z & P_x - iP_y \\ P_x + iP_y & -P_z \end{bmatrix}$ (c)

$$H = \begin{bmatrix} m_e c^2 & 0 & cP_z & c(P_x+iP_y) \\ 0 & m_e c^2 & c(P_x+iP_y) & -cP_z \\ cP_z & c(P_x+iP_y) & -m_e c^2 & 0 \\ c(P_x+iP_y) & -cP_z & 0 & -m_e c^2 \end{bmatrix}$$

$$i\hbar\frac{\partial}{\partial t}\begin{pmatrix}\psi_1\\\psi_2\\\psi_3\\\psi_4\end{pmatrix} = \begin{pmatrix} m_e c^2 & 0 & -i\hbar c\,\partial_z & -i\hbar c(\partial_x - i\partial_y) \\ 0 & m_e c^2 & -i\hbar c(\partial_x + i\partial_y) & i\hbar c\,\partial_z \\ -i\hbar c\,\partial_z & -i\hbar c(\partial_x - i\partial_y) & -m_e c^2 & 0 \\ -i\hbar c(\partial_x + i\partial_y) & i\hbar c\,\partial_z & 0 & -m_e c^2 \end{pmatrix}\begin{pmatrix}\psi_1\\\psi_2\\\psi_3\\\psi_4\end{pmatrix}$$

(b) $\hat{v} = c\bar{\alpha}$, $\hat{v} = \dfrac{dr}{dt} = \dfrac{1}{i\hbar}[\bar{r}, \bar{H}]$

$\hat{v} = \dfrac{1}{i\hbar}[\bar{r}, c\bar{\alpha}\cdot\bar{p} + \beta m_e c^2]$

$$[r_i, p_j] = i\hbar\,\delta_{ij}$$

$$\hat{v}_i = \frac{1}{i\hbar}\left[r_i, \sum_j c\alpha_j p_j\right] = \frac{c}{i\hbar}\sum_j \alpha_j [r_i, p_j]$$

$$\Rightarrow \hat{v}_i = \frac{c}{i\hbar}\sum_j \alpha_j (i\hbar\delta_{ij}) = c\alpha_i$$

$$\Rightarrow \boxed{\hat{v} = c\hat{\alpha}}$$

(c) $$i\hbar\frac{\partial\psi}{\partial t} = -i\hbar c\bar{\alpha}\cdot\nabla\psi + \beta m_e c^2\psi \qquad ①$$

$$-i\hbar\frac{\partial\psi^\dagger}{\partial t} = i\hbar c(\nabla\psi^\dagger)\cdot\bar{\alpha} + m_e c^2\psi^\dagger\beta \qquad ②$$

$$\psi^\dagger \times ① - ② \times \psi$$

$$\Rightarrow i\hbar\left(\psi^\dagger\frac{\partial\psi}{\partial t} - \frac{\partial\psi^\dagger}{\partial t}\psi\right) = -i\hbar c\left[\psi^\dagger\bar{\alpha}\cdot\nabla\psi + (\nabla\psi^\dagger)\cdot\bar{\alpha}\psi\right]$$

$$\Rightarrow \boxed{i\hbar\frac{\partial}{\partial t}(\psi^\dagger\psi) = -i\hbar c\,\nabla\cdot(\psi^\dagger\bar{\alpha}\psi)}$$

$$\Rightarrow \boxed{\frac{\partial}{\partial t}(\psi^\dagger\psi) + \nabla(c\,\psi^\dagger\bar{\alpha}\psi) = 0}$$

$$\Rightarrow \boxed{P + \bar{\nabla}\cdot\bar{J} = 0}$$

Ans 4) (a) $[\bar{L}, \bar{H}] = i\hbar c(\bar{\alpha} \times \bar{p})$

$\bar{L} = \bar{r} \times \bar{p}$, $L_i = \epsilon_{ijk} r_j P_k$

$[L_i, H] = [\epsilon_{ijk} r_j P_k, c\bar{\alpha}\cdot\bar{p} + \beta m_e c^2]$

$= c \sum_m \alpha_m [\epsilon_{ijk} r_j P_k, P_m]$

$= c\,\epsilon_{ijk} \sum_m \alpha_m [r_j, P_m] P_k$

$[r_j, P_m] = i\hbar \delta_{jm}$

$[L_i, H] = c\,\epsilon_{ijk} \sum_m \alpha_m (i\hbar \delta_{jm}) P_k = i\hbar c \sum_k \epsilon_{ikj} \alpha_j P_k$

$= i\hbar c(\bar{\alpha} \times \bar{p})_i$

$\Rightarrow [L, H] = i\hbar c(\bar{\alpha} \times \bar{p}) \neq 0$

$\downarrow$

angular momentum not conserved

(b) $\Sigma = \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix}$ $\qquad \Sigma_x = -i\alpha_y\alpha_z$

$\Sigma_x = -i \begin{pmatrix} 0 & \sigma_y \\ \sigma_y & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{pmatrix} = -i \begin{pmatrix} \sigma_y\sigma_z & 0 \\ 0 & \sigma_y\sigma_z \end{pmatrix}$

$\sigma_j \sigma_k = \delta_{jk} I + i\,\epsilon_{jki}\sigma_i$, $\sigma_y \sigma_z = i\sigma_i$

$\Sigma_x = \begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}$

(c) $S = (\hbar/2)\Sigma$

$$\Rightarrow S^2 = S_x^2 + S_y^2 + S_z^2 = \frac{\hbar^2}{4}(\Sigma_x^2 + \Sigma_y^2 + \Sigma_z^2)$$

$$= \frac{3\hbar^2}{4} I_4$$

$\Sigma_z = \begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}$ has eigenvalues ± 1

$\Rightarrow$ eigenvalues of S are $\pm\hbar/2$

(d) $\bar{J} = \bar{L} + \bar{S}$

$$[S_i, H] = \frac{\hbar}{2}\left[\Sigma_i, c\sum_j \alpha_j p_j\right] = \frac{\hbar c}{2}\sum_j [\Sigma_i, \alpha_j] p_j$$

evaluating $[\Sigma_i, \alpha_j]$:

$$\begin{pmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}\begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix} - \begin{pmatrix} 0 & \sigma_j \\ \sigma_j & 0 \end{pmatrix}\begin{pmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix} = \begin{pmatrix} 0 & [\sigma_i, \sigma_j] \\ [\sigma_i, \sigma_j] & 0 \end{pmatrix}$$

$$\Rightarrow [\Sigma_i, \alpha_j] = 2i\,\epsilon_{ijk}\alpha_k$$

$$[S_i, H] = -i\hbar c(\bar{\alpha} \times \bar{p})_i$$

$$\Rightarrow [S, H] = -i\hbar c[\bar{\alpha} \times \bar{p}]$$

$$[\bar{J}, \bar{H}] = [\bar{L}, \bar{H}] + [\bar{S}, \bar{H}] = i\hbar c(\bar{\alpha} \times \bar{p}) - i\hbar c(\bar{\alpha} \times \bar{p})$$

$$= 0$$

(e) $[\bar{H}, \bar{S}\cdot\bar{p}] = \sum_i [H, S_i p_i] = \sum_i \left([H, S_i]p_i + S_i[H, p_i]\right)$

now $[\bar{H}, p_i] = 0$

$[\bar{H}, \bar{S}\cdot\bar{p}] = \sum_i [H, S_i]p_i = -\sum_i [S_i, H]p_i$

$\Rightarrow [\bar{H}, \bar{S}\cdot\bar{p}] = i\hbar c(\bar{\alpha}\times\bar{p})\cdot\bar{p} = 0$

$\Rightarrow \boxed{[\bar{H}, \bar{S}\cdot\bar{p}] = 0}$

(f) $\hat{n} = \dfrac{\bar{\Sigma}\cdot\hat{p}}{|\bar{p}|}$, $\hat{n}^2 = \dfrac{(\Sigma\cdot\hat{p})^2}{|\bar{p}|^2}$

$(\bar{\Sigma}\cdot\bar{A})(\bar{\Sigma}\cdot\bar{B}) = \bar{A}\cdot\bar{B}\,I + i\Sigma(\bar{A}\times\bar{B})$, $\bar{A} = \bar{B} = \bar{p}$

$(\bar{\Sigma}\cdot\bar{p})^2 = |\bar{p}|^2 I + i\bar{\Sigma}\cdot(\bar{p}\times\bar{p}) = |\bar{p}|^2 I$

$\hat{n}^2 = \dfrac{|P|^2}{|P|^2} = I_4$

its eigenvalues are ± 1

$\Psi_+ = \begin{pmatrix} \chi_+ \\ c_1\chi_+ \end{pmatrix}$ $\Psi_- = \begin{pmatrix} \chi_- \\ c_2\chi_- \end{pmatrix}$

such that :

$(\bar{\sigma}\cdot\hat{n})\chi_\pm = \pm\chi_\pm$

Ans 5 (a) $i\hbar \frac{\partial \psi(\bar{r},t)}{\partial t} = H\psi(r,t) = (c\bar{\alpha}\cdot\bar{p} + \beta m_e c^2)\psi$

$$E\psi = (c\bar{\alpha}\cdot\bar{p} + \beta m_e c^2)\psi$$

$$\psi = \begin{pmatrix} v \\ \omega \end{pmatrix} \qquad \bar{\alpha} = \begin{pmatrix} 0 & \bar{\sigma} \\ \bar{\sigma} & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$$

$$E\begin{pmatrix} v \\ \omega \end{pmatrix} = \begin{pmatrix} m_e c^2 & c\bar{\sigma}\cdot\bar{p} \\ c\bar{\sigma}\cdot\bar{p} & -m_e c^2 \end{pmatrix}\begin{pmatrix} v \\ \omega \end{pmatrix}$$

This gives: $(E - m_e c^2)v - c(\bar{\sigma}\cdot\bar{p})\omega = 0$

$-c(\bar{\sigma}\cdot\bar{p})v + (E + m_e c^2)\omega = 0$

(b) $\omega = \frac{c(\bar{\sigma}\cdot\bar{p})}{E + m_e c^2} v$

$$\Rightarrow (E - m_e c^2)v - c(\bar{\sigma}\cdot\bar{p})\left[\frac{c(\bar{\sigma}\cdot\bar{p})}{E + m_e c^2} v\right] = 0$$

$$\Rightarrow (E - m_e c^2)(E + m_e c^2)v - c^2(\bar{\sigma}\cdot\bar{p})^2 v = 0$$

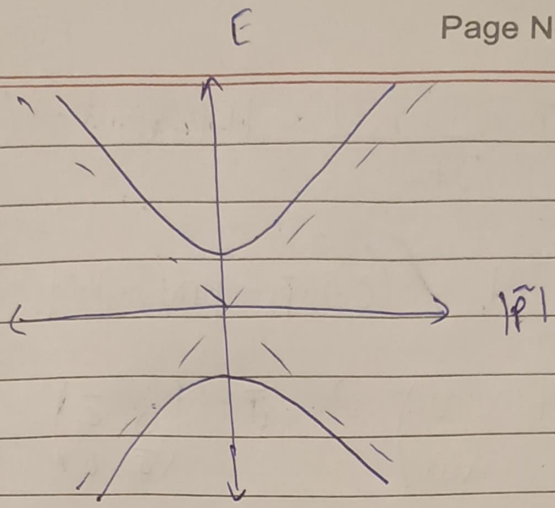
identity : $(\bar{\sigma}\cdot\bar{A})(\bar{\sigma}\cdot\bar{B}) = \bar{A}\cdot\bar{B} + i\bar{\sigma}\cdot(\bar{A}\times\bar{B})$

$\Rightarrow (\bar{\sigma}\cdot\bar{p})^2 = |p|^2 = p^2$ $\qquad A = B = p$

$$\Rightarrow (E^2 - m_e^2 c^4 - p^2c^2)v = 0$$

$$E^2 = p^2c^2 + m_e^2c^4 \Rightarrow E(p) = \pm\sqrt{p^2c^2 + m_e^2c^4}$$

$$\frac{E}{m_e c^2} = \pm\sqrt{1 + |\hat{p}|^2}$$



(c) $E_+ = +\sqrt{p^2c^2 + m_e^2c^4}$

chose upper spinor v.

$$\omega = \frac{c(\bar{\sigma}\cdot\bar{p})}{E_+ + m_e c^2}\, v$$

$$\psi^{(1)} = N\begin{pmatrix} 1 \\ 0 \\ \dfrac{cp_z}{E_+ + m_e c^2} \\ \dfrac{c(p_x + ip_y)}{E_+ + m_e c^2} \end{pmatrix} \qquad \psi^{(2)} = N\begin{pmatrix} 0 \\ 1 \\ \dfrac{c(p_x - ip_y)}{E_+ + m_e c^2} \\ \dfrac{-cp_z}{E_+ + m_e c^2} \end{pmatrix}$$

For $E_- = -\sqrt{p^2c^2 + m_e^2c^4}$:

$$\psi^{(3)} = \begin{pmatrix} \dfrac{-cp_z}{E_+ + m_e c^2} \\ \dfrac{-c(p_x + ip_y)}{E_+ + m_e c^2} \\ 1 \\ 0 \end{pmatrix} \qquad \psi^{(4)} = \begin{pmatrix} \dfrac{-c(p_x - ip_y)}{E_+ + m_e c^2} \\ \dfrac{cp_z}{E_+ + m_e c^2} \\ 0 \\ 1 \end{pmatrix}$$

(d) $\psi^\dagger \psi = 1$ for normalization

$$1 = |N|^2 \left[1 + \frac{p^2 c^2}{(E_+ + m_e c^2)^2} \right]$$

$$\Rightarrow |N|^2 \left[1 + \frac{E_+ - m_e c^2}{E_+ + m_e c^2} \right] \Rightarrow N = \sqrt{\frac{E_+ + m_e c^2}{2E_+}}$$

$$\langle v \rangle = c\bar{\alpha} \quad , \quad \psi = \begin{pmatrix} v \\ w \end{pmatrix}$$

$$\langle v \rangle = \psi^\dagger (c\bar{\alpha}) \psi = c(v^\dagger \; w^\dagger) \begin{pmatrix} 0 & \sigma \\ \sigma & 0 \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix}$$

$$\langle v \rangle = c(v^\dagger \bar{\sigma} w + w^\dagger \bar{\sigma} v)$$

$$\Rightarrow \boxed{\langle v \rangle = \bar{p} c^2 / E}$$

$$\langle S \rangle = \psi^\dagger \frac{\hbar}{2} \begin{pmatrix} \bar{\sigma} & 0 \\ 0 & \bar{\sigma} \end{pmatrix} \psi = \boxed{\frac{\hbar}{2} N^2 (v^\dagger \bar{\sigma} v + w^\dagger \bar{\sigma} w)}$$

taking $\psi^{(1)}$, $\bar{p} = p_z \hat{k}$, $\langle S \rangle = (0, 0, \hbar/2)$

ANS 6) a) $\begin{pmatrix} m_e c^2 + V & c\bar{\sigma}\cdot\bar{p} \\ c\bar{\sigma}\cdot\bar{p} & -m_e c^2 + V \end{pmatrix} \begin{pmatrix} \psi \\ \chi \end{pmatrix} = E \begin{pmatrix} \psi \\ \chi \end{pmatrix}$

$$E = m_e c^2 + \epsilon$$

$$c(\bar{\sigma}\cdot\bar{p})\psi + (-2m_e c^2 + V - \epsilon)\chi = 0$$

$$\Rightarrow \chi = \frac{c(\bar{\sigma}\cdot\bar{p})}{2m_e c^2 + \epsilon - V} \psi$$

For $\epsilon, V << 2m_e c^2$:

$$\chi \approx \frac{1}{2m_e c}\left\{1 - \frac{\epsilon - V}{2m_e c^2}\right\}(\bar{\sigma}\cdot\bar{p})\psi$$

Substitute χ into top row eq^n:

$$\epsilon\psi = V\psi + c(\bar{\sigma}\cdot\bar{p})\left[\frac{1}{2m_e c}\left(1 - \frac{\epsilon - V}{2m_e c^2}\right)(\bar{\sigma}\cdot\bar{p})\right]\psi$$

$$\epsilon\psi = \left[V + \frac{(\bar{\sigma}\cdot\bar{p})^2}{2m_e} - \frac{1}{4m_e^2c^2}(\bar{\sigma}\cdot\bar{p})(\epsilon - V)(\bar{\sigma}\cdot\bar{p})\right]\psi$$

$$(\bar{\sigma}\cdot\bar{p})^2 = p^2$$

$\Rightarrow$ first two terms give: $\left(\frac{p^2}{2m_e} + V\right)\psi = \epsilon\psi$

$$[\bar{p}, \bar{V}] = -i\hbar\nabla V$$

(i) relativistic KE correction:

$\epsilon - V \approx \frac{p^2}{2m_e}$, gives $\frac{-p^4}{8m_e^3c^2}$

(ii) Spin-Orbit Coupling: $\frac{\hbar}{8m_e^2c^2}\nabla^2 V$

(iii) Darwin term: $\frac{\hbar^2}{8m_e^2c^2}\nabla^2 V$

(b) Zeeman effect

$\bar{B} = \bar{\nabla} \times \bar{A}$

$p \to \pi = \bar{p} + e\bar{A}$

$$\begin{pmatrix} m_e c^2 & c\bar{\sigma}\cdot\bar{\pi} \\ c\bar{\sigma}\cdot\bar{\pi} & -m_e c^2 \end{pmatrix}\begin{pmatrix} \psi \\ \chi \end{pmatrix} = (m_e c^2 + \epsilon)\begin{pmatrix} \psi \\ \chi \end{pmatrix}$$

$\epsilon << 2m_e c^2$

$$c(\bar{\sigma}\cdot\bar{\pi})\psi - 2m_e c^2 \chi \approx 0 \Rightarrow \chi \approx \frac{\bar{\sigma}\cdot\bar{\pi}}{2m_e c}\psi$$

$$\Rightarrow \epsilon\psi = \epsilon(\bar{\sigma}\cdot\bar{\pi})\left(\frac{\bar{\sigma}\cdot\bar{\pi}}{2m_e c}\right)\psi = \frac{(\bar{\sigma}\cdot\bar{\pi})^2}{2m_e}\psi$$

evaluate $(\bar{\sigma}\cdot\bar{\pi})^2$ using $(\bar{\sigma}\cdot\bar{u})(\bar{\sigma}\cdot\bar{v})$
$= \bar{u}\cdot\bar{v} + i\bar{\sigma}\cdot(\bar{u}\times\bar{v})$:

$$(\bar{\sigma}\cdot\bar{\pi})^2 = \bar{\pi}^2 + i\bar{\sigma}\cdot(\bar{\pi}\times\bar{\pi})$$

$$\bar{\pi}\times\bar{\pi} = e(\bar{p}\times\bar{A} + \bar{A}\times\bar{p})$$

$= -ie\hbar B$ (applying to test f^n)

$$\rightarrow (\bar{\sigma}\cdot\bar{\pi})^2 = (\bar{p}+e\bar{A})^2 + e\hbar\,\bar{\sigma}\cdot\bar{B}$$

$$\epsilon\psi = \left[\frac{(\bar{p}+e\bar{A})^2}{2m_e} + \frac{e\hbar}{2m_e}\bar{\sigma}\cdot\bar{B}\right]\psi$$

$\bar{S} = (\hbar/2)\bar{\sigma}$, $g = 2$

$$\frac{e\hbar}{2m_e}\bar{\sigma}\cdot\bar{B} = \frac{e}{m_e}\left(\frac{\hbar\bar{\sigma}}{2}\right)\cdot\bar{B} = \frac{ge}{2m_e}S\cdot B \Rightarrow \boxed{\left[\frac{(\bar{p}+e\bar{A})^2}{2m_e} + g\frac{e}{2m_e}\bar{S}\cdot\bar{B}\right]\psi(r) = \epsilon\psi(r)}$$